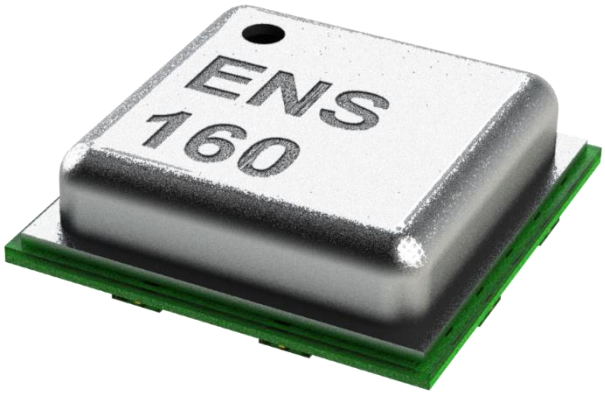
ScioSense™

ENS160



Digital Metal Oxide Multi-Gas Sensor

ENS160 Datasheet

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ScioSense

Digital Metal-Oxide Multi-Gas Sensor

The ENS160 is a digital multi-gas sensor solution, based on metal oxide (MOX) technology with four MOX sensor elements. Each sensor element has independent hotplate control to detect a wide range of gases e.g. volatile organic compounds (VOCs) including ethanol, toluene, as well as hydrogen and nitrogen dioxide with superior selectivity and accuracy. For indoor air quality applications, the ENS160 supports intelligent algorithms to digitally process raw sensor measurements on-chip. These algorithms calculate CO_2-equivalents, TVOC, air quality indices (AQIs) and perform humidity and temperature compensation, as well as baseline management – all on chip! Moreover, a development option is available to digitally output raw sensor measurements from each sensor element for customization. The LGA-packaged device includes an SPI or I^2C slave interface with separate VDDIO to communicate with a main host processor. The ENS160 is a proven and maintenance-free technology, designed for high volume and reliability.

Key Features & Benefits

TrueVOC™ air quality detection with industry-leading purity and stability, providing multiple outputs e.g. eCO_2[1], TVOC and AQIs[2] in compliance with worldwide IAQ[3]-signal standards

Independent sensor heater control for highest selectivity (e.g. to ethanol, toluene, acetone, NO_2) and outstanding background discrimination

Immunity to siloxanes and humidity[4]

Hassle-free on-chip heater drive control and data processing – no need for external libraries – no mainboard-CPU performance impacts

Interrupt on threshold for low-power applications

Wide operating ranges: temperature: -40 to +85°C; humidity: 5 to 95%[5]; V_{DD}: 1.71 to 1.98V; V_{DDIO} 1.71 to 3.6V

Applications

- Building Automation / Smarthome / HVAC[6]
  - Indoor air quality detection
  - Demand-controlled ventilation
  - Smart thermostats
- Home appliances
  - Cooker hoods
  - Air cleaners / purifiers
- IoT devices

Properties

- Small-3 x 3 x 0.9mm LGA package
- Design-flexibility through standard, fast and fast mode plus I^2C- and SPI-interfaces with separate VDDIO up to 3.6V
- T&R packaged, reflow-solderable[7]

[1] eCO_2 = equivalent CO_2 values for compatibility with HVAC ventilation standards
[2] AQI = Air Quality Index
[3] IAQ = Indoor Air Quality
[4] T/RH compensation via external T/RH-input
[5] Non-condensing
[6] HVAC = Heat, Ventilation and Air Conditioning
[7] See section "Soldering Information" for further details

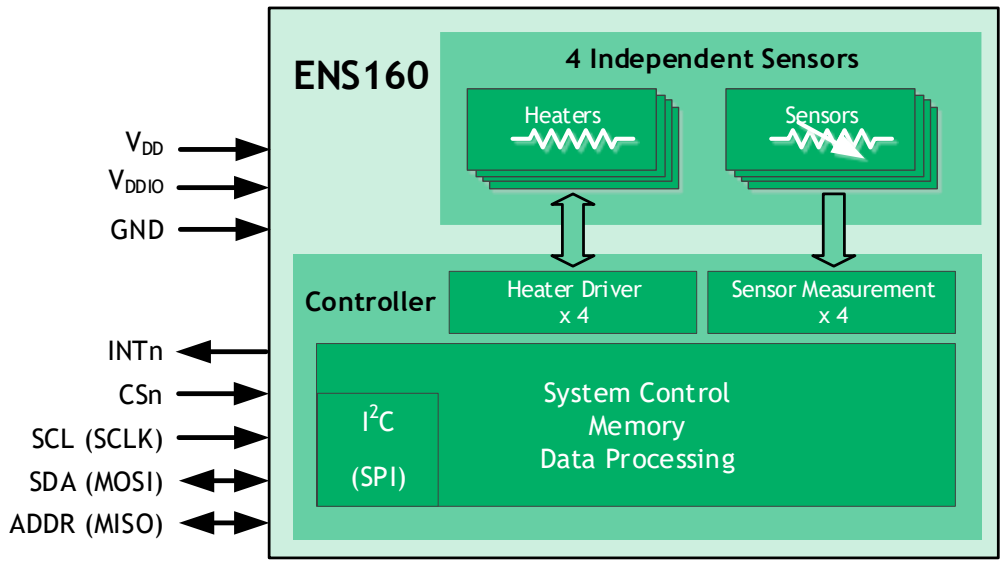
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1 Block Diagram

The ENS160 digital multi-gas sensor consists of four independent heaters and gas sensor elements, based on metal oxide (MOX) technology and a controller as shown in the functional block diagram below.

Figure 1: Functional Blocks



The *Heater Driver* controls the sensor operating modes and provides power to the *heaters* of each individual sensor element. During operation the heater driver regulates the heaters to their individual set-points.

The *Sensor Measurement* block determines the value of the sensor resistance for each individual sensor element.

The *System Control* block processes the resistance values internally to output calculated TVOC, CO_2-equivalents, AQIs and further signals on the digital interface.

The ENS160 includes a standard 2-wire digital *I^2C interface* (SCL, SDA) or 4-wire digital *SPI interface* (SCLK, MOSI, MISO, CSn) for communication to the main host processor.

On-chip memory is used to store calibration values.

2 Pin Assignment

Figure 2: Pin Diagram

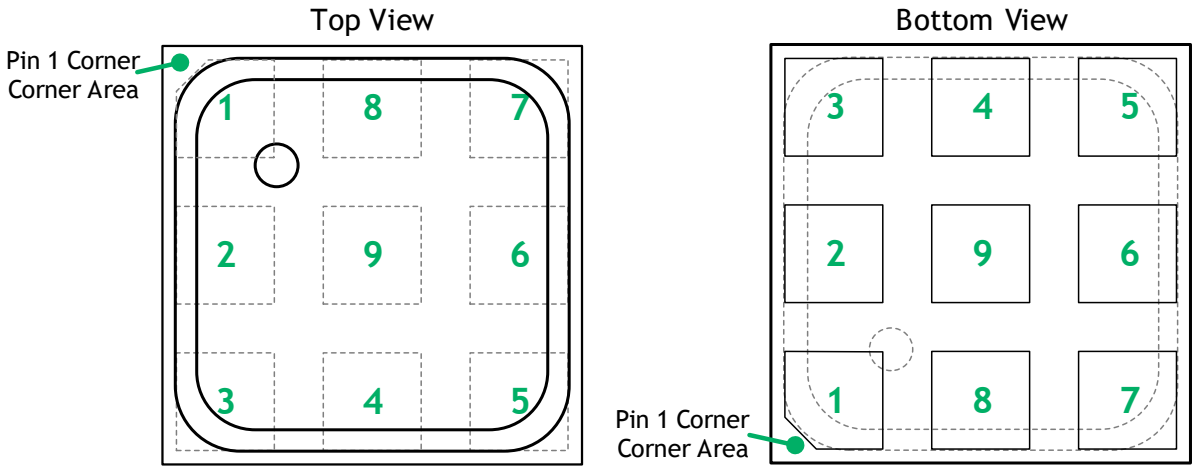


Table 1: Pin Description

Pins	Pin Name	Pin Type	Description
1	MOSI / SDA	Input / Output	SPI Master Output Slave Input / I²C Bus Bi-Directional Data
2	SCLK / SCL	Input	SPI Serial Clock / I²C Bus Serial Clock Input
3	MISO / ADDR	Input / Output	SPI Master Input Slave Output / I²C Address Select: I²C ADDR pin high -> 0x53 / ADDR pin low -> 0x52
4	V_{DD}	Supply	Main Supply Voltage
5	V_{DDIO}	Supply	Interface Supply Pins
6	INTn	Output	Interrupt to Host
7	CSn	Input	SPI Interface Select (CSn low -> SPI / CSn high -> I²C)
8, 9	V_{SS}	Supply	Ground Supply Voltage

Also see sections “I²C Operation Circuitry” and “SPI Operation Circuitry” for wiring.

3 Absolute Maximum Ratings

Table 2: Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Units	Comments
Electrical Parameters					
V_{DD}	Supply Voltage	-0.3	1.98	V	
V_{DDIO}	I/O Interface Supply	-0.3	3.6	V	
V_{IO1}	MOSI/SDA, SCLK/SCL	-0.3	3.6	V	
V_{IO2}	MISO/ADDR, INTn, CSn	-0.3	V_{DDIO}+0.3	V	
V_{SS}	Input Ground	-0.3	0.3	V	
I_{SCR}	Input Current (latch-up immunity)	± 100		mA	AEC-Q100-004
Electrostatic Discharge					
ESD_{HBM}	Electrostatic Discharge HBM	± 2000		V	JS-001-2014
ESD_{CDM}	Electrostatic Discharge CDM	± 750		V	JS-002-2014
Operating and Storage Conditions					
MSL	Moisture Sensitivity Level		1		Unlimited floor lifetime
T_{BODY}	Max. Package Body Temperature		260	°C	IPC/JEDEC J-STD-020
T_{STRG}	Storage Temperature	-40	125	°C	
RH_{STRG}	Storage Relative Humidity	5	95	%	Non-condensing
T_{AMB}[1]	Operating Ambient Temperature	-40	85	°C	
RH_{AMB}[1]	Operating Ambient Rel. Humidity	5	95	%	Non-condensing

Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated under Electrical Characteristics is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability and lifetime.

Important Note: The ENS160 is not designed for use in safety-critical or life-protecting applications.

[1] The ENS160 is electrically operable in this range, however its gas sensing performance might vary. Please refer to "Recommended Sensor Operation" for further information.

4 Electrical Characteristics

The following figure details the electrical characteristics of the ENS160.

Table 3: Electrical Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DD}	Positive supply		1.71	1.8	1.98	V
V_{DDIO}	IO Supply Voltage		1.71		3.6	V
I_{DD}	Average[1] Supply Current[2]	DEEPSLEEP (OP_MODE 0x00)[3]		0.01		mA
		IDLE (OP_MODE 0x01)[3]		2	2.5	mA
		STANDARD (OP_MODE 0x02)		24		mA
I_{DD_PK}	Peak Supply Current[4]	STANDARD (OP_MODE 0x02)		65 (<5ms)		mA
V_{IH}	High-level input voltage		$0.7xV_{DDIO}$			V
V_{IL}	Low-level input voltage				$0.3xV_{DDIO}$	V
V_{OH}	High-level output voltage	MISO[5] [I_{OH}=5mA]	$0.8xV_{DDIO}$			V
		INTN [I_{OH}=2mA]	$0.65xV_{DDIO}$			V
V_{OL}	Low-level output voltage	MOSI/SDA, MISO [I_{OL}=5mA]			$0.2xV_{DDIO}$	V
		INTN [I_{OL}=2mA]			$0.35xV_{DDIO}$	V

[1] Averaged over the sequence
[2] Measured at V_{DD}-pin at ambient temperature of 35°C
[3] Not a gas sensing mode
[4] Initial (<5ms) current demand from VDD after the sensor is switched from IDLE (OP-Mode 1) to STANDARD operation (OP_MODE 2)
[5] MOSI/SDA is open drain

5 Air Quality Signal Characteristics

To satisfy a wide range of individual application requirements, the ENS160 offers a series of (indoor) air quality output signals that are derived from various national and international, as well as de-facto standards. Table 4 provides a summary of such signals, with further description in the following sections.

Table 4: Air Quality Signal Output Characteristics

Parameter	Range	Resolution	Unit	Comment
TVOC	0 – 65 000	1	ppb	For requirements outside these specified ranges please contact us
eCO_2	400 – 65 000	1	ppm CO_2 equiv.	
AQI-UBA[1]	1 to 5	1	-	

5.1 TVOC – Total Volatile Organic Compounds

More than 5000 VOCs exist, and they are two to five times more likely to be found indoors than outdoors. Indoor VOCs are various types of hydrocarbons from mainly two sources: bio-effluents, i.e. odors from human respiration, transpiration and metabolism, and building material including furniture and household supplies. VOCs are known to cause eye irritation, headache, drowsiness or even dizziness – all summarized under the term Sick Building Syndrome (SBS). Besides industrial applications, comfort aspects (e.g. temperature), or building protection (humidity), VOCs are the one and only root cause for ventilation.

To group and classify VOCs, regional guidelines and industry-preferences define a series of compounds and mixtures as reference. E.g. ethanol, toluene, acetone, combinations of the various groups of VOCs (e.g. ISO16000-29), and others.

The ENS160 supplies calibration to ethanol for best, most balanced TVOC-results.

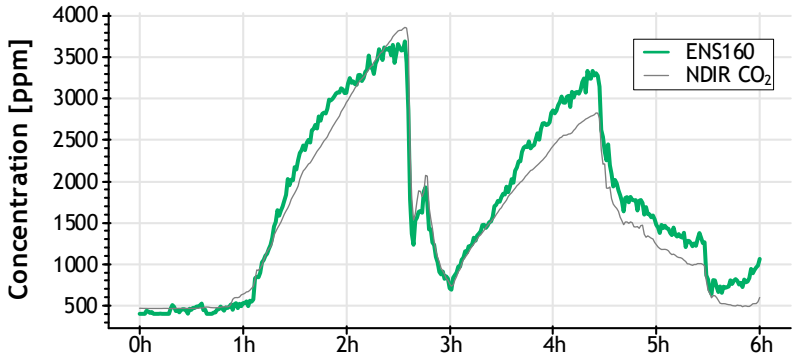
Refer to “Registers” and “DATA_TVOC (Address 0x22)” on how to obtain TVOC-values from the ENS160.

5.2 eCO_2 – Equivalent CO_2

Due to the proportionality between VOCs and -CO_2 generated by humans, CO_2-values historically served as an air quality indicator, reflecting the total amount of VOCs (=TVOCs) produced by human respiration and transpiration. This law (first revealed by Max von Pettenkofer[2] in the 19th century) and the unavailability of suitable VOC measurement technology made CO_2 the surrogate of inhabitant-generated air-pollution in confined living spaces of the past *and* the present, i.e. today’s standard air quality reference for demand-controlled ventilation – as adopted by most HVAC industry standards.

[1] Classified TVOC output signal according to the indoor air quality levels by the German Federal Environmental Agency (UBA, 2007)
[2] Max von Pettenkofer (*1818 – †1901), German chemist and hygienist.

Figure 3: ENS160-based equivalent CO_2 (eCO_2) vs. NDIR-based CO_2 during two meeting sessions



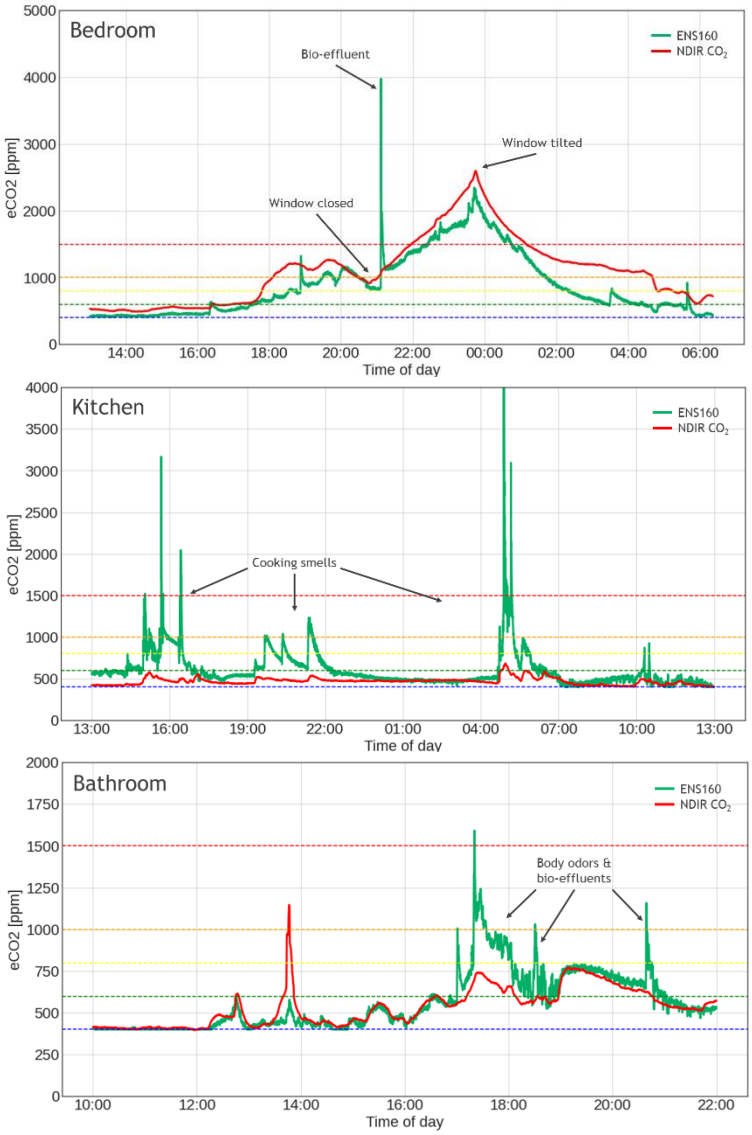
The ENS160 reverses the proportional correlation of VOCs and CO_2, by providing a standardized output signal in ppmCO_2-equivalents from measured VOCs plus hydrogen, thereby adhering to today's CO_2-standards, as shown opposite: ENS160-based equivalent CO_2 estimate vs. CO_2, detected by an NDIR-sensor during two consecutive meeting sessions, interrupted by a lunch-break.

Figure 4: Added value of ENS160's eCO_2 Outputs -where plain CO_2 sensors fail

A key advantage of the ENS160 is the capture of odors and bio-effluents that are completely invisible to CO_2-sensors. The opposite diagrams compare the ENS160's equivalent CO_2 output to an NDIR CO_2 sensor in typical indoor applications:

CO_2 sensors neither detect unpleasant odors and bio-effluents in bedroom or bathroom environments, nor cooking smells in kitchens or restaurants, whereas the ENS160 reliably reports such events.

Proven TrueVOC™ control-algorithms minimize sensor drift and ageing to provide reliable readings over lifetime, thereby making the ENS160's equivalent CO_2 output an affordable solution to complement or substitute real CO_2-based air-quality sensors in the HVAC domain.



The below table shows a typical classification of (equivalent) CO_2 output levels.

Table 5: Interpretation of CO_2 and Equivalent CO_2 Values

Output		Comment / Recommendation
eCO_2 / CO_2	Rating	
>1500	Bad	Heavily contaminated indoor air / Ventilation required
1000 - 1500	Poor	Contaminated indoor air / Ventilation recommended
800 - 1000	Fair	Optional ventilation
600 - 800	Good	Average
400 - 600	Excellent	Target

Example: A CO_2- or eCO_2-controlled ventilation application would invoke its ventilation fan speeds 1, 2 and 3 at the upper three levels "Fair", "Poor" and "Bad", respectively.

See section "Registers" and "DATA_ECO2 (Address 0x24)" on how to obtain equivalent CO_2-values from the ENS160.

5.3 AQI-UBA - Air Quality Index of the UBA[1]

The AQI-UBA air quality index is derived from a guideline by the German Federal Environmental Agency based on a TVOC sum signal. Although a local, German recommendation, this guideline is referenced and adopted by many countries and organizations.

Table 6: Air Quality Index of the UBA (German Federal Environmental Agency)

AQI-UBA		Hygienic Rating	Recommendation	Exposure Limit
#	Rating			
5	Unhealthy	Situation not acceptable	Use only if unavoidable Intensified ventilation recommended	hours
4	Poor	Major objections	Intensified ventilation recommended Search for sources	<1 month
3	Moderate	Some objections	Increased ventilation recommended Search for sources	<12 months
2	Good	No relevant objections	Sufficient ventilation recommended	no limit
1	Excellent	No objections	Target	no limit

Recommendation according to the UBA, Bundesgesundheitsblatt – Gesundheitsforschung Gesundheitsschutz 2007, 50:990–1005, DOI 10.1007/s00103-007-0290-y © Springer Medizin Verlag 2007

See section "Registers" and DATA_AQI (Address 0x21) on how to obtain AQI-values from the ENS160.

[1] UBA = Umweltbundesamt – German Federal Environmental Agency

6 Single Gas Signal Characteristics

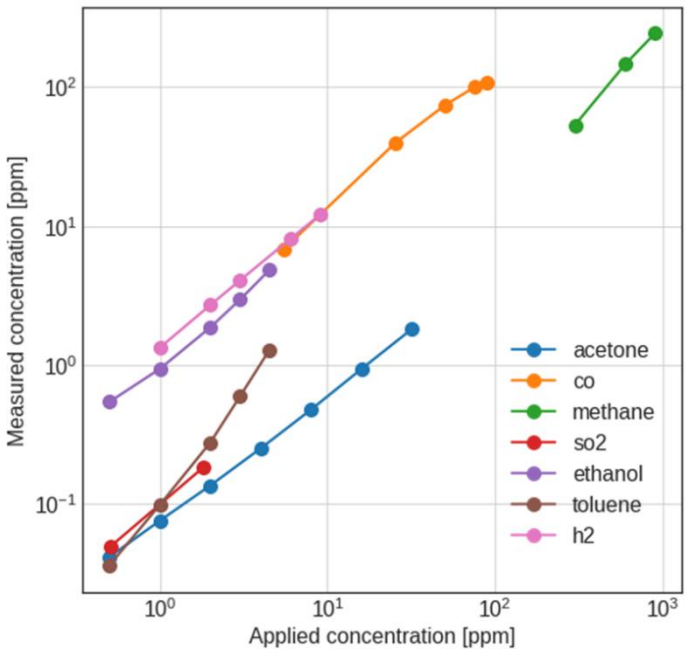


Figure 5: Example Response of the ENS160 to Various Gases

Since metal oxide sensors exhibit a broadband sensitivity to both reducing and oxidizing gases, their raw output signals represent the resulting sum of the entire gas mixture, present. Such sum-signals are beneficial when it comes to wideband TVOC- or AQI-applications, but unsatisfactory for the detection of single gases.

The opposite table shows the response of the ENS160 to a variety of individual gases that can be found indoors.

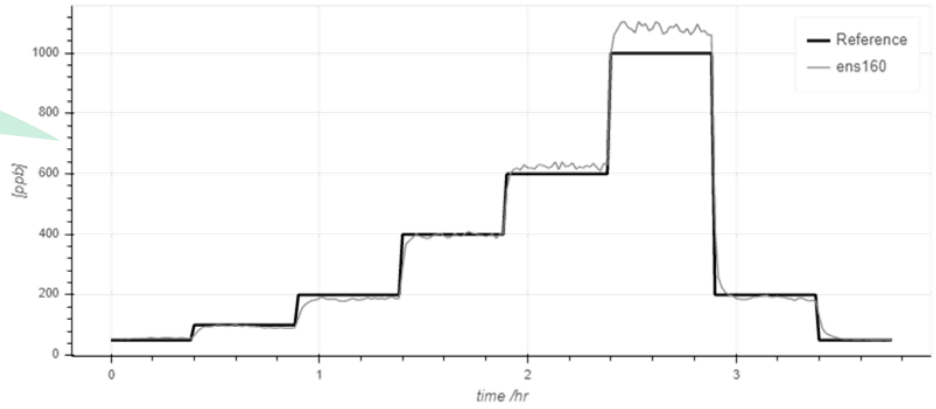
The below table provides a list of selected gases that have been individually characterized.

Table 7: Single Gas Signal Characteristics

Target Gas	Specified Range	Unit	Register	Comment
Ethanol	0 to 6	ppm	DATA_ETOH (0x22) = DATA_TVOC	Dedicated Register
Hydrogen	0 to 10	ppm	$R4_{raw}$ = GPR_READ[6:7]	R_{iraw} = raw resistance values that need to be calibrated to target gas. See text below.
Acetone	0 to 32	ppm	$R4_{raw}$ = GPR_READ[6:7]	
Carbon Monoxide	0 to 90	ppm	$R4_{raw}$ = GPR_READ[6:7]	
Toluene	0 to 32	ppm	$R4_{raw}$ = GPR_READ[6:7]	

Measurement values for individual gases can be obtained from dedicated device registers or calculated from sensor raw resistance values as specified in above table. See sections “Registers” and “Gas Sensor Raw Resistance Signals” for further information.

Figure 6: Example Response of the ENS160 to Ethanol



7 Gas Sensor Raw Resistance Signals

For two of its sensing elements the ENS160 provides individual outputs of raw sensor values.

Table 8: Gas Sensor Raw Resistance Signals

Sensor	Raw Value	Range	Unit	Gen. Purpose Register	Comment
1	$R1_{raw}$	[0..65535]	-	GPR_READ[0:1]	Arbitrary logarithmic units - no resistance values. R_{iraw} require conversion to corresponding resistance value R_{ires} [Ω] (see below)
4	$R4_{raw}$	[0..65535]	-	GPR_READ[6:7]	

Gas sensor raw-values R_{iraw} can be obtained from the ENS160's General Purpose Read Register (GPR_READ) for customer-specific signal post-processing.

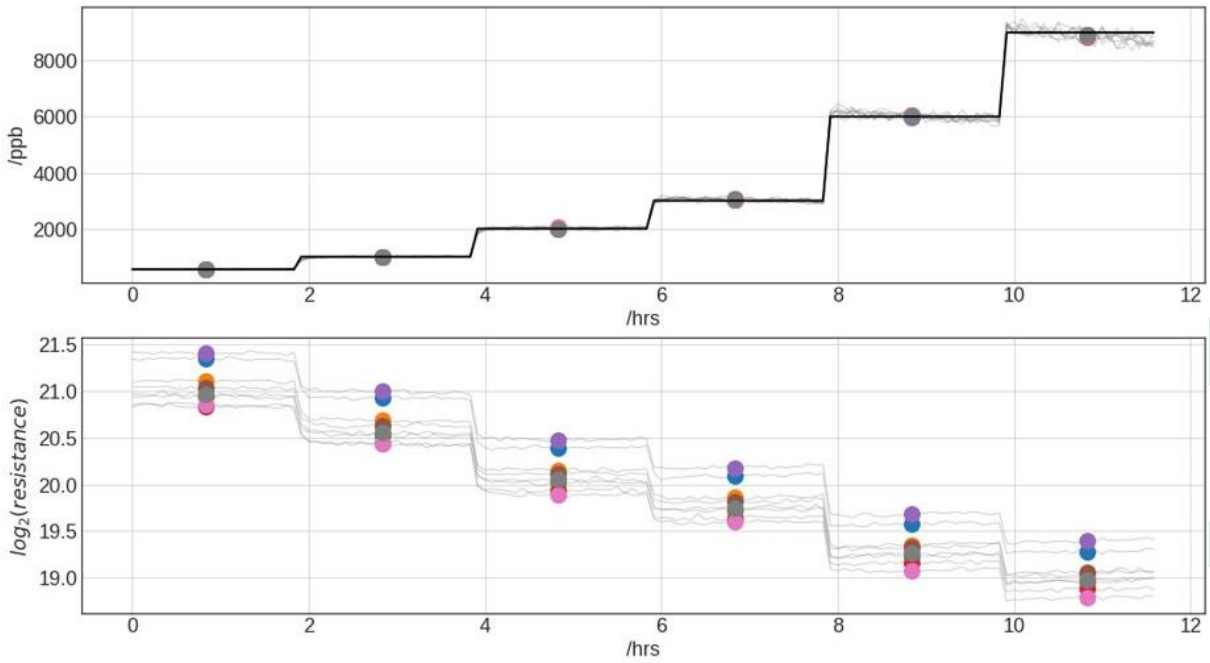
Prior to use R_{iraw} values require conversion to resistance values, using the following formula:

$$R_{ires}[\Omega] = 2^{\frac{R_{iraw}}{2048}}$$

See section "Registers" and GPR_READ (Address 0x48 – 0x4F) on how to obtain AQI-values from the ENS160.

The below figures show the response of eight ENS160s to various hydrogen concentration[1] steps (upper diagram) and the corresponding raw sensor resistance R_{iraw} (lower diagram).

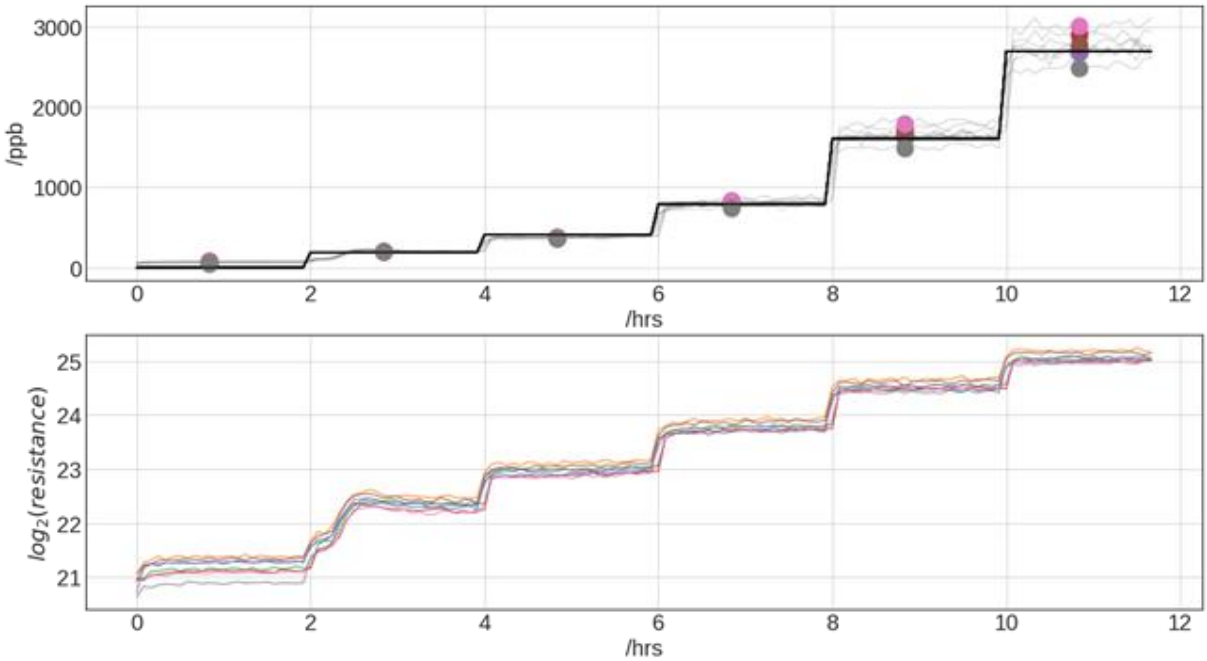
Figure 7: Raw Sensor Signal Response to Hydrogen



[1] Use of the term "Concentration" in ppm (= parts per million) and ppb (= parts per billion) means volume fractions of the respective gases in air: 1 ppm = 1 mL/m³ = 1000 ppb = 1000 µL/m³

The following figures show the response of eight ENS160s to various nitrogen dioxide concentration steps (upper diagram) and the corresponding raw sensor resistance R_{iraw} (lower diagram).

Figure 8: Raw Sensor Signal Response to Nitrogen Dioxide



Note: Due to the nature of sensor raw resistance values, these signals are not conditioned, i.e. not compensated for drift, ageing or cross-sensitivity (interference of background gases including humidity).

8 Signal Conditioning

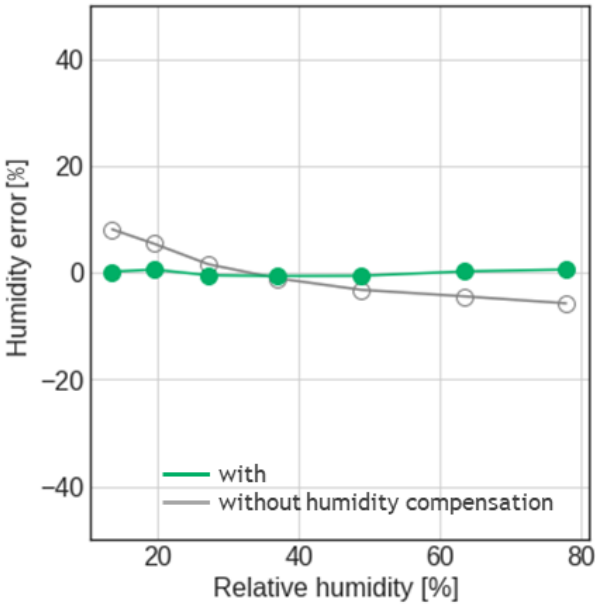
Chemical gas sensors are relative sensors that are susceptible to changes in their chemical and physical environments. Typical drivers are changes of the target gas(es), of the interfering background gas mixture and changes of the physical environment (air pressure, humidity, etc.).

8.1 Baselining

As part of the TrueVOC™ technology the ENS160 deploys an automatic baseline correction, featuring compensation for oxidizing gases such as ozone. It furthermore stores the current baseline value in non-volatile memory to automatically start from the latest valid level of background air after re-powering the device and even after a power outage.

8.2 Humidity Behavior & Compensation

Figure 9: Air Quality Signal with and without Humidity Compensation



For use in normal air quality applications (eCO_2, TVOC, AQI), operated in a relative humidity range between 20 and 80%, the ENS160 does not require external humidity compensation, as the opposite graph shows.

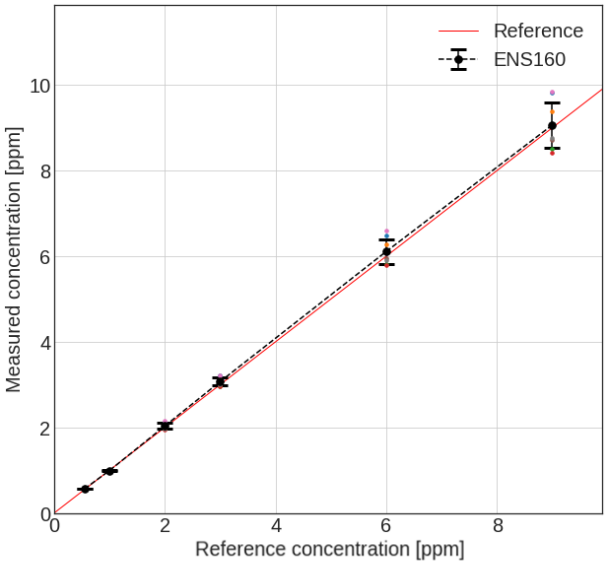
Extreme humidity conditions outside this range (20% - 80%RH) can influence the output signal, especially when very accurate or single gas measurements are required. To overcome such impacts, the ENS160 is equipped with a temperature and humidity compensation algorithm, relying on data from an external temperature- and humidity-sensor (the ENS160 works well with the ScioSense ENS21x family of temperature and humidity sensors as they both share the same signal format), that can be regularly updated to an internal register for processing.

Note: Unless otherwise stated, the humidity compensation discussed in this section works per default for all output signals except for sensor raw signals.

See sections “Registers”, “TEMP_IN” and “RH_IN” for further information.

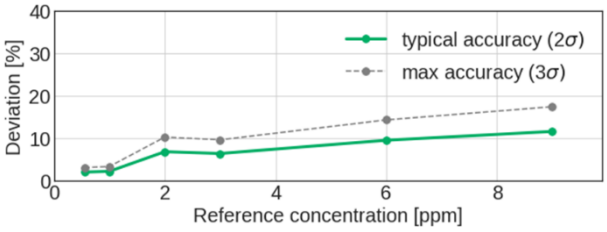
9 Output Signal Accuracy[1]

Figure 10: Output Signal Accuracy for Hydrogen



The ENS160 exhibits an excellent measurement accuracy and device-to-device variation.

The opposite diagrams show the non-linearity of several devices (left) and typical and maximum accuracies (bottom) for various hydrogen concentrations. A typical error of <12% of the measured value can be stated.



10 Initial Start-Up and Warm-Up

Table 9: Initial Start-Up and Warm-Up Timings

Parameter	Maximum Time	Comment
Initial Start-Up	1 hour	See below for further details
Warm-Up	1 minute	

10.1 Initial Start-Up

Initial Start-Up is the time the ENS160 needs to exhibit reasonable air quality readings after its first ever power-on.

The ENS160 sensor raw resistance signals and sensitivities will change upon first power-on. The change in resistance is greatest in the first 48 hours of operation. Therefore, the ENS160 employs a start-up algorithm, allowing eCO_2-, TVOC- and AQI-output signals to be used from first power-on after 1 hour of operation[2].

10.2 Warm-Up

Further to “Initial Start-Up” the conditioning or “Warm-Up” period is the time required to achieve adequate sensor stability before measuring VOCs after idle periods or power-off. Typically, the ENS160 requires 1 minute of warm-up before reasonable air quality readings can be expected[1].

[1] All values have been determined by tests in clean, partially synthetic air in a climate chamber-with stated environmental conditions, suitable reference analytics and sensor preconditioning of at least 24h, which may not reflect real-life environments. Unless otherwise noted, the accuracy statements have been carried out at 25°C and 50% relative humidity.
[2] Slightly reduced signal accuracy may be encountered in early phase, thereafter.

11 Gas Sensor Status and Signal Rating

The status flag is an additional feature assessing the current operational mode and the reliability of the output signals. It aids the application obligation to manage timings efficiently, in particular during initial start-up or after re-powering. Furthermore, a simple signal quality assessment and a system self-check is provided.

Table 10: ENS160 Status and Signal Rating (Validity Flag)

Flag	Meaning	Implementation approach
0	Operating ok	Standard operating mode.
1	Warm-up	During first minute after power-on.
2	Initial Start-up	During first hour after power-on. Only once in a lifetime.
3	No valid output	Signals give unexpected values (very high or very low). Multiple sensors out of range.

See “Validity Flag” in section “DATA_STATUS” for further information.

12 Recommended Sensor Operation

For best performance, the sensor shall be operated in normal indoor air in the range -5 to 60°C (typical: 25°C); relative humidity: 20 to 80%RH (typical: 50%RH), non-condensing with no aggressive or poisonous gases present. Prolonged exposure to environments outside these conditions can affect performance and lifetime of the sensor.

Please also refer to the “ENS160 Design Guidelines and Handling Instructions” for further information on handling and optimal integration of the ENS160. The guidelines in this document must be met for optimal sensor performance and long lifetime.

Important Note: The ENS160 is not designed for use in any safety-critical or life-protecting application.

13 Recommended Sensor Storage

The guidelines under “Recommended Sensor Operation” also apply for sensor storage.

14 Host Communication

The ENS160 is an I²C or SPI Slave device.

If the CSn is held high, the interface behaves as an I²C slave. At power-up the condition of the MISO/ADDR pin is used to determine the LSB of the I²C address. The I²C slave address is 0x52 (MISO/ADDR low) or 0x53 (MISO/ADDR high).

If the CSn pin is asserted (low) the interface behaves as an SPI slave. This condition is maintained until the next Power-on Reset.

Both the SPI and I²C slave interfaces use the same register map for communication.

14.1 I²C Specification

14.1.1 I²C Description

The ENS160 is an I²C slave device with a fixed 7-bit address 0x52 if the MISO/ADDR line is held low at power-up or 0x53 if the MISO/ADDR line is held high.

The I²C interface supports standard (100kbit/s), fast (400kbit/s), and fast plus (1Mbit/s) mode. Details on I²C protocol is according to I²C-bus specifications [UM10204, I²C-bus specification and user manual, Rev. 6, 4 April 2014].

The device applies all mandatory I²C protocol features for slaves: START, STOP, Acknowledge and 7-bit slave address. None of the other optional features (10-bit slave address, general call, software reset or Device ID) are supported, nor are the master features (Synchronization, Arbitration, START byte).

The Host System, as an I²C master, can directly read or write values to one of the registers by first sending the single byte register address. The ENS160 implements “auto increment” which means that it is possible to read or write multiple bytes (e.g. read multiple DATA_X bytes) in a single transaction.

14.1.2 I²C I/O and Timing Information

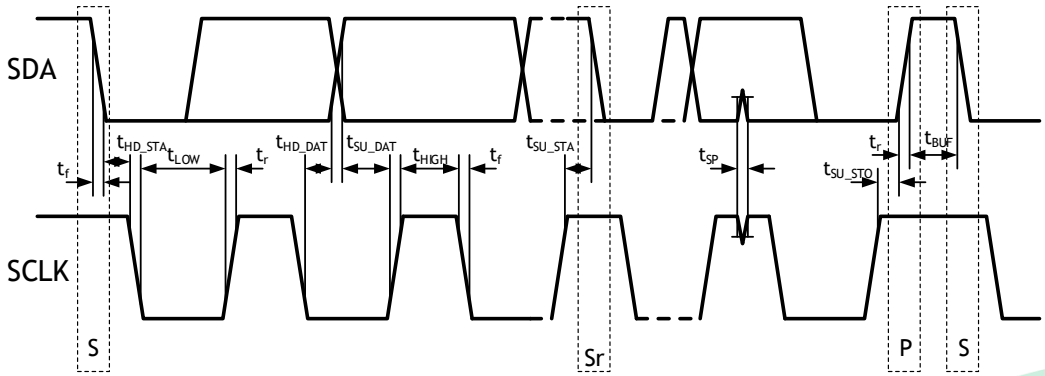
Table 11: ENS160 I²C I/O Parameters

Parameter	Symbol	Standard		Fast		Fast Mode Plus		Unit
		Min	Max	Min	Max	Min	Max	
Low level input voltage	V_{IL}	-0.5	$0.3xV_{DDIO}$	-0.5	$0.3xV_{DDIO}$	-0.5	$0.3xV_{DDIO}$	V
High level input voltage	V_{IH}	$0.7xV_{DDIO}$	2.39	$0.7xV_{DDIO}$	2.39	$0.7xV_{DDIO}$	2.39	V
Hysteresis of Schmitt trigger inputs	V_{hys}	-	-	$0.05xV_{DDIO}$	-	$0.05xV_{DDIO}$	-	V
Low-level output voltage @ 2mA sink current	V_{OL2}	-	-	0	$0.2xV_{DDIO}$	0	$0.2xV_{DDIO}$	V
Low-level output current @ 0.4V	I_{OL}	3		3		20		mA
Output fall time from V_{IHmin} to V_{ILmax}	t_{OF}		250	$20xV_{DDIO}$ / 5.5	250	$20xV_{DDIO}$ / 5.5	250	ns
Input current each I/O pin	I_i	-10	10	-10	10	-10	10	µA

Table 12: ENS160 I²C Timing Parameters[1]

Parameter	Symbol	Standard		Fast		Fast Mode Plus		Unit
		Min	Max	Min	Max	Min	Max	
SCLK clock frequency	f_{SCLK}	0	100	0	400	0	1000	kHz
Hold time (repeated) START condition. After this period, the first clock pulse is generated	t_{HD_STA}	4	-	0.6	-	0.26	-	µs
LOW period of the SCLK clock	t_{LOW}	4.7	-	1.3	-	0.5	-	µs
HIGH period of the SCLK clock	t_{HIGH}	4.0	-	0.6	-	0.26	-	µs
Set-up time for a repeated START condition	t_{SU_STA}	4.7	-	0.6	-	0.26	-	µs
Data set-up time	t_{SU_DAT}	250	-	100 [2]	-	50 [2]	-	ns
Data hold-time	t_{HD_DAT}	0 [3]	3.45 [4]	0 [3]	0.9 [4]	0 [3]	-	µs
Rise time of SDA and SCLK signals	t_r	-	1000	20	300	20	120	ns
Fall time of SDA and SCLK signals	t_f	-	300	$20xV_{DDIO}$ / 5.5	300	$20xV_{DDIO}$ / 5.5	120	ns
Set-up time for STOP condition	t_{SU_STO}	4.0	-	0.6	-	0.26	-	µs
Bus free time between a STOP and START condition	t_{BUF}	4.7	-	1.3	-	0.5	-	µs
Capacitive load for each bus line	C_b	-	400	-	400	-	550	pF
Noise margin at the LOW level	V_{nL}	$0.1xV_{DDIO}$	-	$0.1xV_{DDIO}$	-	$0.1xV_{DDIO}$	-	V
Noise margin at the HIGH level	V_{nH}	$0.2xV_{DDIO}$	-	$0.2xV_{DDIO}$	-	$0.2xV_{DDIO}$	-	V

Figure 11: Definition of I²C Timing Parameters



[1] All values referred to V_{IHmin} and V_{ILmax} levels
[2] A fast mode I²C bus device can be used in Standard mode I²C bus system, but the requirement t_{SU_DAT} >= 250ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal. If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line t_{rmax}. t_{SU_DAT}= 1000 + 250 = 1250ns (according to standard mode I²C bus specification) before the SCL line is released.
[3] This device internally provides a hold time of at least 300ns for the SDA signal to bridge the undefined region of the falling edge of the SCL
[4] The maximum t_{HD_DAT} has only to be met if the device does not stretch the LOW period (t_{LOW}) of the SCLK signal

14.1.3 I²C Read Operation

After the START condition, in the first transaction:

- The I²C Master sends the 7-bit slave address and 0 into the R/W bit (the byte sent would be 0xA4 or 0xA6 dependent on the power-up value of MISO/ADDR).
- The I²C Master then sends the address of the first register to read.

Then either after a RESTART condition (i.e. STOP followed by START)

- The I²C Master sends the 7-bit slave address and 1 into the R/W bit (the byte sent would be 0xA5 or 0xA7 dependent on the power-up value of MISO/ADDR).
- The I²C Master then reads 1-n data bytes from sequential registers (if valid) until the transaction is concluded with a STOP condition.

Figure 12: I²C Read Operation

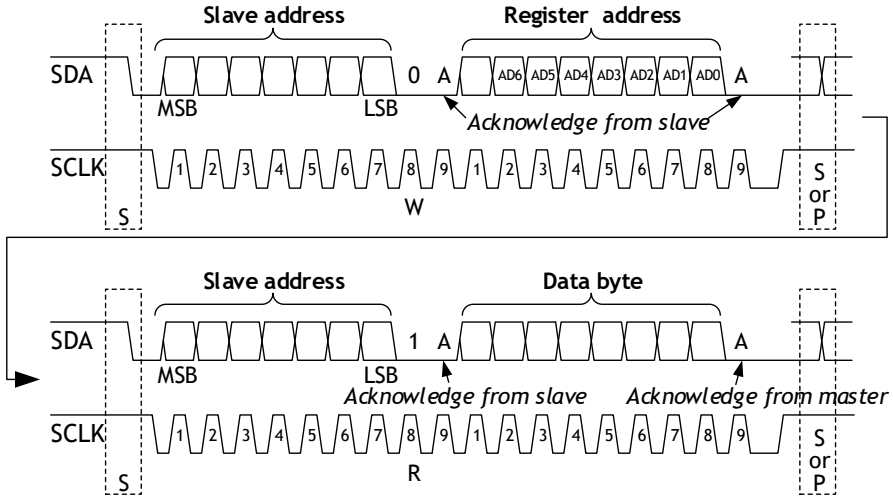
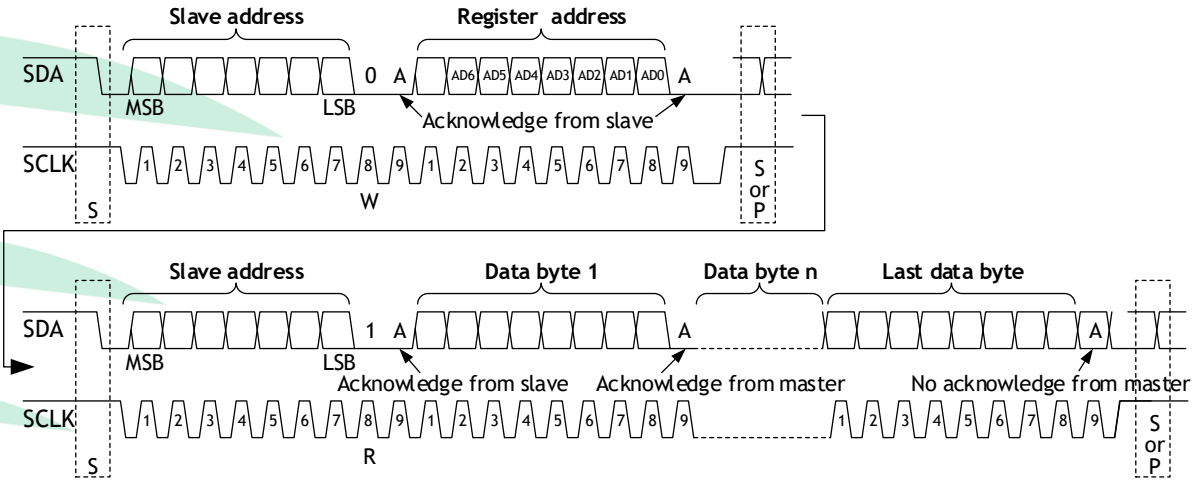


Figure 13: I²C Auto-Increment Read Operation



14.1.4 I²C Write Operation

After the START condition, in a single continuous transaction:

- The I²C Master sends the 7-bit slave address and 0 into the R/W bit (the byte sent would be 0xA4 or 0xA6 dependent on the power-up value of MISO/ADDR).
- The I²C Master then sends the address of the first register to write.
- The I²C Master then sends 1-n data bytes which are written into sequential registers (if valid) until the transaction is concluded with a STOP condition.

Figure 14: I²C Write Operation

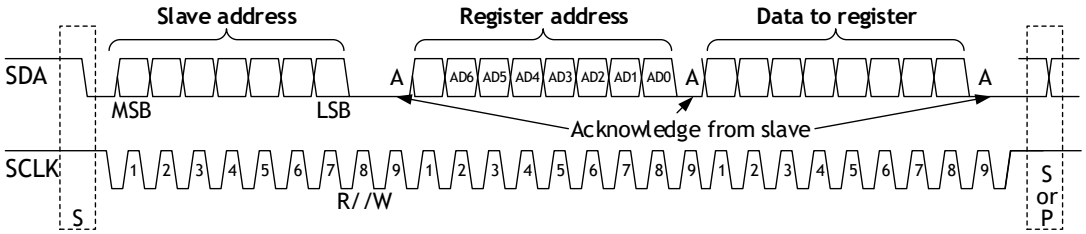
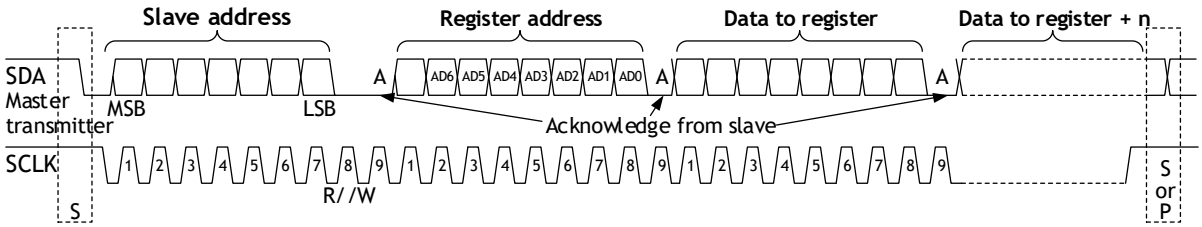


Figure 15: I²C Auto-Increment Write Operation



14.2 SPI Specification

14.2.1 SPI Description

The SPI interface is a slave bus operating up to 10MHz clock-frequency.

It shares pins with the I²C interface. SPI is selected and SPI transfer initiated by asserting the CSn line low. Once the CSn line has been asserted low the ENS160 will not accept I²C transactions until the next Power-On Reset.

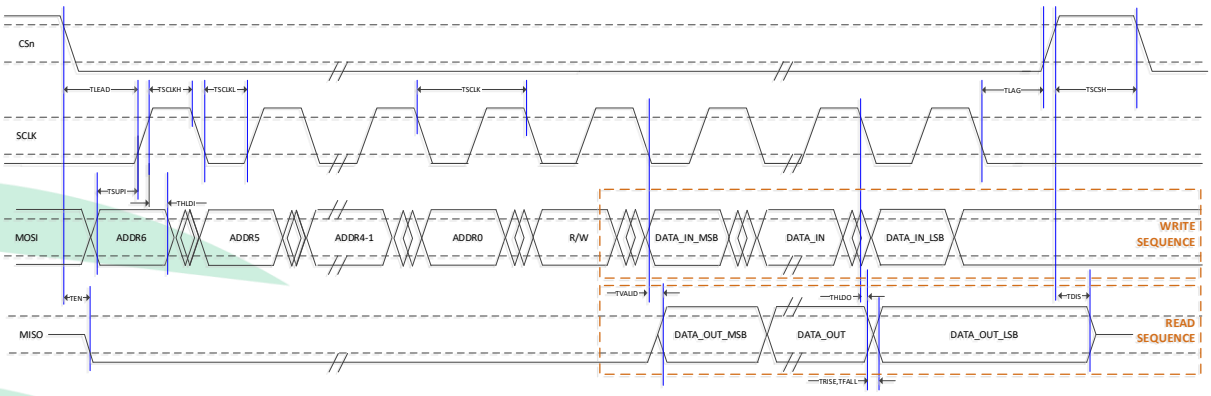
Data is clocked in on the rising edge of SCLK; most significant bit first.

14.2.2 SPI Timing Information

Table 13: SPI Timings

Parameter	Symbol	Condition	Min	Typ	Max	Unit
SPI Clock (SCLK) Frequency	FSCLK				10	MHz
CSn falling to MISO Enabled	TEN	25pF load			20	ns
CSn rising to MISO Disable	TDIS	25pF load			20	ns
MOSI Setup Time before SCLK	TSUPI		15			ns
MOSI hold time after rising SCLK	THLDI		15			ns
CSn low to first rising SCLK	TLEAD		20			ns
Last SCLK low to CSn high	TLAG		20			ns
SCLK High Time	TSCLKH		40			ns
SCLK Low Time	TSCLKL		40			ns
SCLK falling to MISO Valid	TVALID	25pF load			40	ns

Figure 16 SPI Timings Reference



14.2.3 SPI Read Operation

During a Read operation, data is clocked out on the falling edge of SCLK so it is stable for the following riding edge.

MISO stays in high impedance mode until the device is selected (CSn low). Data on MISO is only valid on a Read operation.

A transaction starts with the target address and R/W control bit in the first byte followed by the read or write data.

In a Read operation Auto-increment of the address enables multiple registers to be read in sequence. CSn de-asserting (to high) terminates the Read sequence.

A Read SPI frame is composed as follows:

Table 14: Read SPI Frame

Byte	Bit	Name	Description
0	7:1	AD[6:0]	On MOSI: Address of the register to Read
0	0	RW	On MOSI: 1: bytes are to be read, starting from AD[6:0].
1	7:0	RDATA[7:0]	Output on MISO; MOSI ignored
n	7:0	RDATA[7:0]	Output on MISO; MOSI ignored

14.2.4 SPI Write Operation

In a Write operation, the address does not Auto-increment. Multiple writes can be performed by alternating Address and Data bytes. CSn de-asserting (to high) terminates the Write sequence.

A Write SPI frame is composed as follows:

Table 15: Write SPI Frame

Byte	Bit	Name	Description
0	7:1	AD[6:0]	On MOSI: Address of the register to Write
0	0	RW	On MOSI: 0: bytes are to be Written, at AD[6:0].
1	7:0	WDATA[7:0]	Input on MOSI; MISO Dummy Data
even	7:1	AD[6:0]	On MOSI: Address of the register to Write
even	0	RW	On MOSI: 0: bytes are to be Written, at AD[6:0].
odd	7:0	WDATA[7:0]	Input on MOSI; MISO Dummy Data

15 Operation

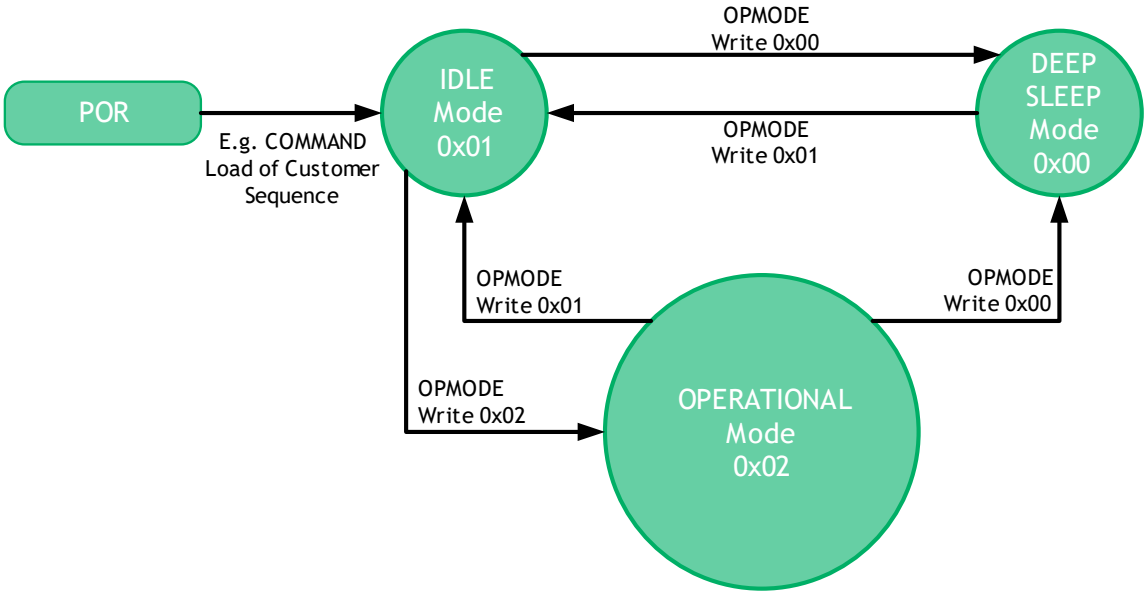
At power-up, the ENS160 configures itself from a reset state and prepares for commands over the serial bus via either I^2C or SPI Protocols.

The default state is OPMODE 0x01, which is an IDLE condition that enables ENS160 so that it may respond to several commands. In this mode it is not operating as a gas sensor.

OPMODE 0x00 is a very low power standby state, called DEEP SLEEP.

Active OPMODEs are described further in the OPMODE Register section.

Figure 17 Orchestration of Operational Modes



Note: When the active gas sensing OPMODE (e.g. 0x02 = STANDARD) is running, new data is notified either via the interrupt (INTn) or by polling the DATA_STATUS register. The output of the gas sensing OPMODEs are presented in the DATA_XXX registers which can be read at any time.

16 Registers

This section describes the registers of the ENS160 which enable the host system to

- Identify the Device and version information
- Configure the ENS160 and set the operating mode
- Read back STATUS information, the calculated gas concentrations and Air Quality Indices

16.1 Register Overview

Note that some registers are spread over multiple addresses. For example, PART_ID at address 0 is spread over 2 addresses (its "Size" is 2). Registers are stored in little endian so the LSB of PART_ID is at address 0 and the MSB of PART_ID is at address 1.

Table 16: Register Overview

Address	Name	Size	Access	Description
0x00	PART_ID	2	Read	Device Identity 0x01, 0x60
0x10	OPMODE	1	Read / Write	Operating Mode
0x11	CONFIG	1	Read / Write	Interrupt Pin Configuration
0x12	COMMAND	1	Read / Write	Additional System Commands
0x13	TEMP_IN	2	Read / Write	Host Ambient Temperature Information
0x15	RH_IN	2	Read / Write	Host Relative Humidity Information
0x17 - 0x1F	-	1	-	Reserved
0x20	DEVICE_STATUS	1	Read	Operating Mode
0x21	DATA_AQI	1	Read	Air Quality Index
0x22	DATA_TVOC	2	Read	TVOC Concentration (ppb)
0x24	DATA_ECO2	2	Read	Equivalent CO_2 Concentration (ppm)
0x26	-	2	-	Reserved
0x28	-	2	-	Reserved
0x2A	-	2	Read	Reserved
0x2C - 0x2F	-	1	-	Reserved
0x30	DATA_T	2	Read	Temperature used in calculations
0x32	DATA_RH	2	Read	Relative Humidity used in calculations
0x34 - 0x37	-	1	-	Reserved
0x38	DATA_MISR	1	Read	Data Integrity Field (optional)
0x40	GPR_WRITE[0:7]	8	Read/Write	General Purpose Write Registers
0x48	GPR_READ[0:7]	8	Read	General Purpose Read Registers

16.2 Detailed Register Description

16.2.1 PART_ID (Address 0x00)

This 2-byte register contains the part number in little endian of the ENS160.

The value is available when the ENS160 is initialized after power-up.

Table 17: Register PART_ID

Address 0x00			PART_ID	
Bits	**Field Name**	**Default**	**Access**	**Field Description**
0:7	PART_ID_LSB	0x60	read	Lower Byte of Part ID
8:15	PART_ID_MSB	0x01	read	Upper Byte of Part ID

16.2.2 OPMODE (Address 0x10)

This 1-byte register sets the Operating Mode of the ENS160. The Host System can write a new OPMODE at any time.

Any current operating mode will terminate and the new operating mode will start.

Table 18: Register OPMODE

Address 0x10			OPMODE	
Bits	**Field Name**	**Default**	**Access**	**Field Description**
7:0		0x00	R/W	Operating mode: 0x00: DEEP SLEEP mode (low power standby) 0x01: IDLE mode (low-power) 0x02: STANDARD Gas Sensing Modes

In DEEP SLEEP mode, ENS160 has limited functionality but will respond to an OPMODE write.

Idle Mode is intended for configuration before running an active sensing mode.

0x02 (STANDARD) is an active gas sensing operating mode to indicate the levels of air quality or for specific gas detection.

16.2.3 CONFIG (Address 0x11)

This 1-byte register configures the action of the INTn pin which allows the ENS160 to signal to the host system that particular data is available.

The INTn pin can be (de-)asserted (polarity configurable) when ENS160 updates GPR_Read registers, or when it updates DATA registers, or when a certain threshold is reached (set through COMMAND mode).

A typical setting 0x23 would enable an active low interrupt (no pull-up required) when new output data is available in the DATA registers.

Table 19: Register CONFIG

Address 0x11				CONFIG
Bits	Field Name	Default	Access	Field Description
7	-	0b0	-	Reserved
6	INTPOL	0b0	R/W	INTn pin polarity: 0: Active low (Default) 1: Active high
5	INT_CFG	0b0	R/W	INTn pin drive: 0: Open drain 1: Push / Pull
4	-	0b0	-	Reserved
3	INTGPR	0b0	R/W	INTn pin asserted when new data is presented in the General Purpose Read Registers
2	-	0b0	-	Reserved
1	INTDAT	0b0	R/W	INTn pin asserted when new data is presented in the DATA_XXX Registers
0	INTEN	0b0	R/W	INTn pin is enabled for the functions above

16.2.4 COMMAND (Address 0x12)

This 1-byte register allows some additional commands to be executed on the ENS160. This register can be written at any time, but commands will only be actioned in IDLE mode (OPMODE 0x01).

The COMMAND register allows multiple interactions with the system where data needs to be passed between the user/host and the ENS160.

Typically, a request for data (e.g. GetHWVer, GetFWVer) will result in the requested data being placed in the General Purpose READ Registers and an input of data (e.g. set alarm threshold) would first be stored in the General Purpose WRITE Registers at address 0x40-47.

Below is a list of valid commands for the ENS160.

Table 20: Register COMMAND

Address 0x12				COMMAND
Bits	Field Name	Default	Access	Command
7:0	Command	0x00	R/W	0x00: ENS160_COMMAND_NOP 0x0E: ENS160_COMMAND_GET_APPVER – Get FW Version 0xCC: ENS160_COMMAND_CLRGPR Clears GPR Read Registers

16.2.4.1 ENS160_COMMAND_GET_APPVER

After issuing ENS160_COMMAND_GET_APPVER, the firmware version of the ENS160 will be placed in General Purpose Registers GPR_READ0 and GPR_READ1. The NEWGPR bit in DATA_STATUS will be set and the INTn asserted if configured to react to NEWGPR.

Table 21: GPR_READ Settings for ENS160_COMMAND_GET_APPVER Command

Register	7	6	5	4	3	2	1	0
GPR_READ0	Release				Version			
GPR_READ1	Sub-Version							

16.2.4.2 ENS160_COMMAND_CLRGPR

After issuing ENS160_COMMAND_CLRGPR all GPR Read registers are cleared.

16.2.5 TEMP_IN (Address 0x13)

This 2-byte register allows the host system to write ambient temperature data to ENS160 for compensation. The register can be written at any time. TEMP_IN_LSB should be written first as the update is recognized on a write to TEMP_IN_MSB.

Table 22: Register TEMP_IN

Address 0x13				TEMP_IN
Bits	Field Name	Default	Access	Field Description
0:7	TEMP_IN _LSB	0x00	R/W	Lower Byte of TEMP_IN
8:15	TEMP_IN_MSB	0x00	R/W	Upper Byte of TEMP_IN

The format of the temperature data is the same as the format used in the ENS21x (family of ScioSense temperature and humidity sensors) as shown below:

Table 23: Format of Temperature Data

Byte 0x14								Byte 0x13							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
TEMP_IN Integer Part (Kelvin)								TEMP_IN Fractions							

The ENS160 required input format is: temperature in Kelvin * 64
(with Kelvin = Celsius + 273.15).

Example: For 25°C the input value is calculated as follows: (25 + 273.15) * 64 = 0x4A8A.

16.2.6 RH_IN (Address 0x15)

This 2-byte register allows the host system to write relative humidity data to ENS160 for compensation. The register can be written at any time. RH_IN_LSB should be written first as the update is recognized on a write to RH_IN_MSB.

Table 24: Register RH_IN

Address 0x15				RH_IN
Bits	Field Name	Default	Access	Field Description
0:7	RH_IN _LSB	0x00	R/W	Lower Byte of RH_IN
8:15	RH_IN_MSB	0x00	R/W	Upper Byte of RH_IN

The format of the relative humidity data is the same as the format used in the ENS21x as shown below:

Table 25: Format of Relative Humidity Data

Byte 0x16								Byte 0x15							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
RH_IN Integer Part (%)								RH_IN Fractions							

The ENS160 required input format is: relative humidity in %rH * 512.

Example: For 50% rH the input value is calculated as follows: 50 * 512 = 0x6400.

16.2.7 DATA_STATUS (Address 0x20)

This 1-byte register indicates the current STATUS of the ENS160.

Table 26: Register DATA_STATUS

Address 0x20				DATA_STATUS
Bits	**Field Name**	**Default**	**Access**	**Field Description**
7	STATAS	0b0	-	High indicates that an OPMODE is running
6	STATER	0b0	R	High indicates that an error is detected. E.g. Invalid Operating Mode has been selected.
5	-	0b0	R	Reserved
4	-	0b0	R	Reserved
2-3	VALIDITY FLAG	0b00	R	Status 0: Normal operation 1: Warm-Up phase 2: Initial Start-Up phase 3: Invalid output
1	NEWDAT	0b0	R	High indicates that a new data is available in the DATA_x registers. Cleared automatically at first DATA_x read.
0	NEWGPR	0b0	R	High indicates that a new data is available in the GPR_READx registers. Cleared automatically at first GPR_READx read.

During operation, Bit 6 (STATER) of DATA_STATUS is asserted if an error has occurred.

The meaning of the errors may be different, depending on the operation being undertaken.

Further information regarding the error can be read from the GPR_READ registers.

16.2.8 DATA_AQI (Address 0x21)

This 1-byte register reports the calculated Air Quality Index according to the UBA.

Table 27: Register DATA_ AQI

Address 0x21				DATA_AQI
Bits	**Field Name**	**Default**	**Access**	**Field Description**
0:2	AQI_UBA	0x01	R	Air Quality Index according to UBA [1..5]
3:7	Reserved	0x00	R	Reserved

See section “AQI-UBA – Air Quality Index of the UBA” for further information.

16.2.9 DATA_TVOC (Address 0x22)

This 2-byte register reports the calculated TVOC concentration in ppb.

Table 28: Register DATA_TVOC

Address 0x22				DATA_TVOC
Bits	Field Name	Default	Access	Field Description
0:7	TVOC_LSB	0x00	R	Lower Byte of DATA_TVOC
8:15	TVOC _MSB	0x00	R	Upper Byte of DATA_TVOC

See section “TVOC – Total Volatile Organic Compounds” for further information.

16.2.10 DATA_ECO2 (Address 0x24)

This 2-byte register reports the calculated equivalent CO_2-concentration in ppm, based on the detected VOCs and hydrogen.

Table 29: Register DATA_ECO2

Address 0x24				DATA_ECO2
Bits	Field Name	Default	Access	Field Description
0:7	ECO2_LSB	0x00	R	Lower Byte of DATA_ECO2
8:15	ECO2_MSB	0x00	R	Upper Byte of DATA_ECO2

See section “eCO_2 – Equivalent CO_2” for further information.

16.2.11 DATA_ETOH (Address 0x22)

This 2-byte register reports the calculated ethanol concentration in ppb. For dual use the DATA_ETOH register is a virtual mirror of the ethanol-calibrated DATA_TVOC register.

Table 30: Register DATA_ETH

Address 0x22				DATA_ETOH
Bits	Field Name	Default	Access	Field Description
0:7	ETH_LSB	0x00	R	Lower Byte of DATA_ETH
8:15	ETH_MSB	0x00	R	Upper Byte of DATA_ETH

16.2.12 DATA_T (Address 0x30)

This 2-byte register reports the temperature used in its calculations (taken from TEMP_IN, if supplied).

Table 31: Register DATA_T

Address 0x30				DATA_T
Bits	Field Name	Default	Access	Field Description
0:7	DATA_T _LSB	0x8A	R	Lower Byte of DATA_T
8:15	DATA_T _MSB	0x4A	R	Upper Byte of DATA_T

The format of the temperature data is the same as the format used in the ENS21x.

Table 32: Format of Temperature Data

Byte 0x30								Byte 0x31							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
TEMP_IN Integer Part (Kelvin)								TEMP_IN Fractions							

The DATA_T storage format is: temperature in Kelvin * 64
(with Kelvin = Celsius + 273.15).

Example: For a stored DATA_T value of 0x4A8A the temperature in °C is calculated as follows: 0x4A8A / 64 - 273.15 = 25°C.

See section “TEMP_IN” for further information.

16.2.13 DATA_RH (Address 0x32)

This 2-byte register reports the relative humidity used in its calculations (taken from RH_IN if supplied).

Table 33: Register DATA_RH

Address 0x32				DATA_RH
Bits	**Field Name**	**Default**	**Access**	**Field Description**
0:7	DATA_RH _LSB	0x00	R	Lower Byte of DATA_RH
8:15	DATA_RH_MSB	0x64	R	Upper Byte of DATA_RH

The format of the relative humidity data is the same as the format used in the ENS21x.

Table 34: Format of Relative Humidity Data

Byte 0x32								Byte 0x33							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
RH_IN Integer Part (%)								RH_IN Fractions							

The DATA_RH storage format is: relative humidity in %rH * 512.

Example: For a stored DATA_RH value of 0x6400 the relative humidity in % is calculated as follows: 0x6400 / 512 = 50%rH.

See section “RH_IN” for further information.

16.2.14 DATA_MISR (Address 0x38)

This 1-byte register reports the calculated checksum of the previous DATA_ read transaction (of n-bytes). It can be read as a separate transaction, if required, to check the validity of the previous transaction. The value should be compared with the number calculated by the Host system on the incoming Data.

Table 35: Register DATA_MISR

Address 0x38				DATA_MISR
Bits	**Field Name**	**Default**	**Access**	**Field Description**
0:7	DATA_MISR	0x00	R	Calculated checksum of the previous transaction

Example: C-code to calculate MISR on the received DATA, to compare with DATA_MISR:

```
// The polynomial used in the CRC computation in DATA_MISR
//                      76543210 bit weight factor
#define POLY 0x1D // 0b00011101 = x^8+x^4+x^3+x^2+x^0 (x^8 is implicit)
// The hardware register DATA_MISR is updated with every read from a
// register in the range 0x20 to 0x37, using a CRC polynomial (POLY).
// For every register read, call `misr_update()` to keep the software
// variable `misr` in sync with the hardware register.
static uint8_t misr = 0; // Mirror of DATA_MISR (0 is hardware default)
uint8_t misr_update(uint8_t data) {
  uint8_t misr_xor= ( (misr<<1) ^ data) & 0xFF;
  if( misr&0x80==0 )
    misr= misr_xor;
  else
    misr= misr_xor ^ POLY;
}
// Typically, when an I2C/SPI transaction is completed, read DATA_MISR,
// and compare it with the software `misr`. They should equal. If not
// there is a CRC error: one or more bytes were corrupted in the transfer.
uint8_t misr_set(void) {
  return misr;
}
// Once the CRC is wrong, or transactions have been executed without
// calling update() the software `misr` is out of sync with DATA_MISR.
// Read DATA_MISR and call `misr_set()` to bring back in sync.
void misr_set(uint8_t * val) {
  misr= val;
}
```

16.2.15 GPR_WRITE (Address 0x40)

This 8-byte register is used by several functions for the Host System to pass data to the ENS160. Writes to these registers are not valid when the ENS160 is in DEEP SLEEP or during a low power portion of an operating mode. Writes should only be done during IDLE mode (OPMODE 0x01).

Table 36: Register GPR_WRITE

Address 0x40				GPR_WRITE0-7	
Address	Bits	Field Name	Default	Access	Field Description
0x40	0:7	GPR_WRITE0	0x00	R/W	General Purpose WRITE Register 0
0x41	0:7	GPR_WRITE1	0x00	R/W	General Purpose WRITE Register 1
0x42	0:7	GPR_WRITE2	0x00	R/W	General Purpose WRITE Register 2
0x43	0:7	GPR_WRITE3	0x00	R/W	General Purpose WRITE Register 3
0x44	0:7	GPR_WRITE4	0x00	R/W	General Purpose WRITE Register 4
0x45	0:7	GPR_WRITE5	0x00	R/W	General Purpose WRITE Register 5
0x46	0:7	GPR_WRITE6	0x00	R/W	General Purpose WRITE Register 6
0x47	0:7	GPR_WRITE7	0x00	R/W	General Purpose WRITE Register 7

16.2.16 GPR_READ (Address 0x48)

This 8-byte register is used by several functions for the ENS160 to pass data to the Host System. When New GPR_DATA is available the NEW_GPR bit of the DATA_STATUS register will be set and the INTn pin asserted (if configured).

Table 37: Register GPR_READ

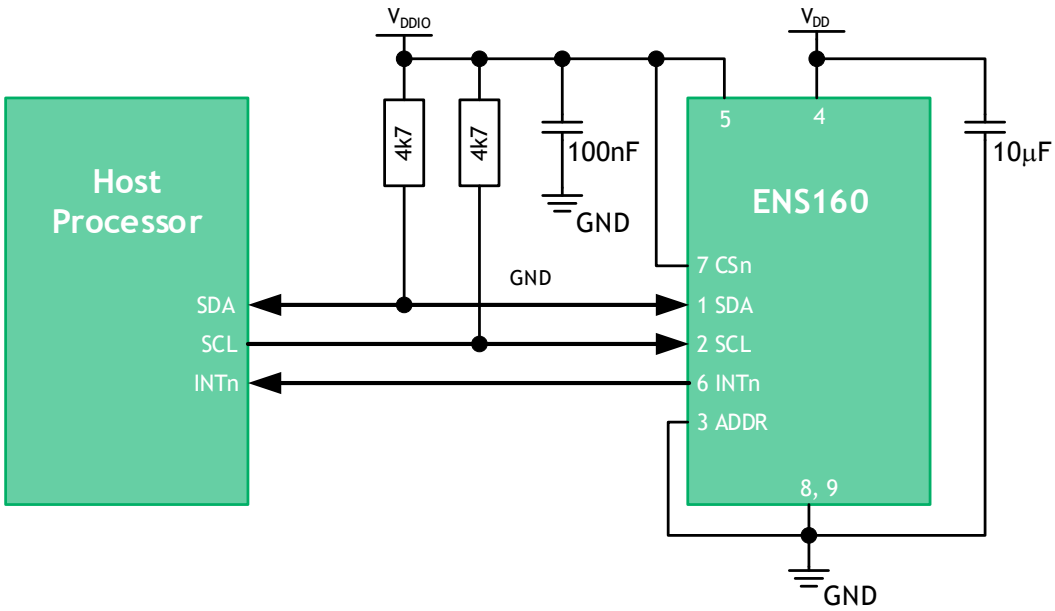
Address 0x48					GPR_READ0-7
Address	**Bits**	**Field Name**	**Default**	**Access**	**Field Description**
0x48	0:7	GPR_READ0	0x00	R	General Purpose READ Register 0
0x49	0:7	GPR_READ1	0x00	R	General Purpose READ Register 1
0x4A	0:7	GPR_READ2	0x00	R	General Purpose READ Register 2
0x4B	0:7	GPR_READ3	0x00	R	General Purpose READ Register 3
0x4C	0:7	GPR_READ4	0x00	R	General Purpose READ Register 4
0x4D	0:7	GPR_READ5	0x00	R	General Purpose READ Register 5
0x4E	0:7	GPR_READ6	0x00	R	General Purpose READ Register 6
0x4F	0:7	GPR_READ7	0x00	R	General Purpose READ Register 7

17 Application Information

17.1 I²C Operation Circuitry

The recommended application circuit for the ENS160 I²C interface operation is shown below:

Figure 18: Recommended Application Circuit (I²C Operation)



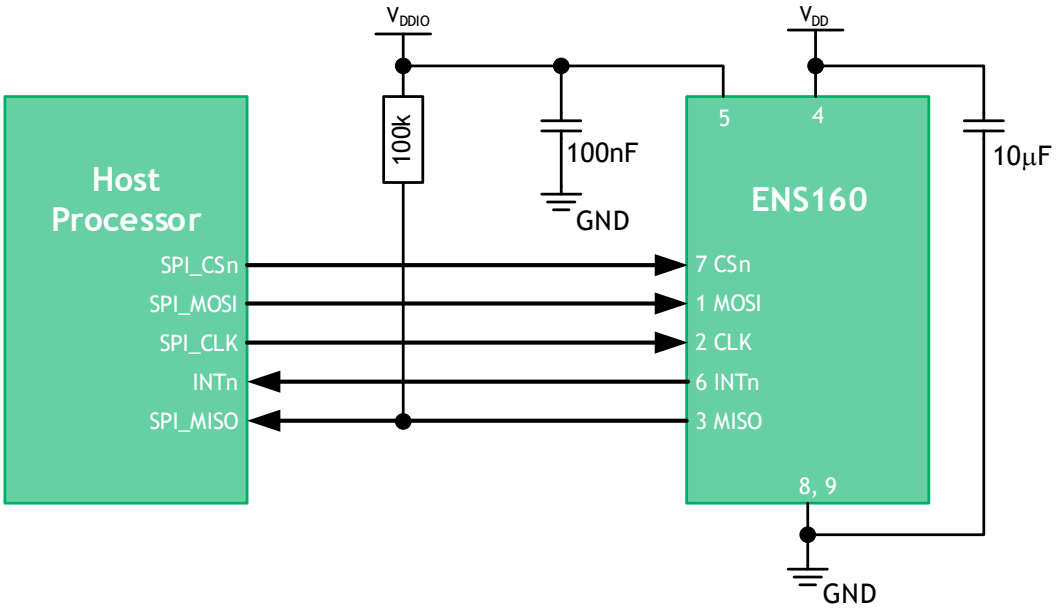
Note(s):

1. CSn must be pulled high (directly to V_{DDIO}) to ensure I²C interface is selected
2. MISO/ADDR should be pulled low or high to specify the LSB of the address
3. Pull-up resistors
   The above recommendation for pull-up resistance values applies to I²C standard mode only. Pull-up resistors for SCL and SDA are assumed to be part of the host system and should be selected dependent on the intended I²C data rate and individual bus architecture.
4. Decoupling capacitor must be placed close to the V_{DD} (Pin 4) and V_{DDIO} (Pin 5) supply pins of the ENS160

17.2 SPI Operation Circuitry

The recommended application circuit for the ENS160 for SPI interface is shown below:

Figure 19: Recommended Application Circuit (SPI Operation)



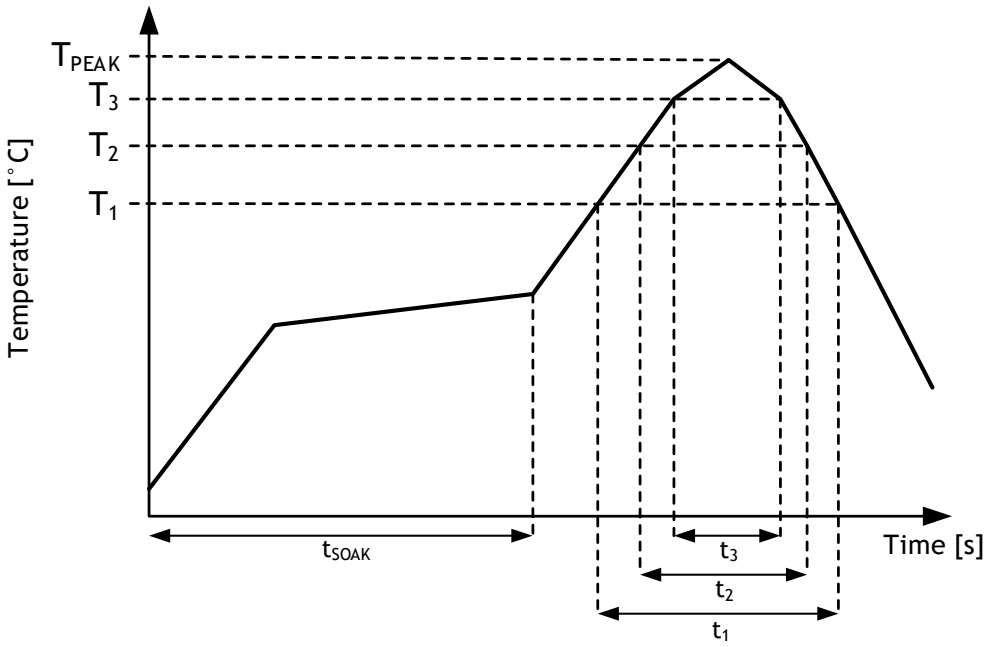
Note(s):

1. Weak pull-up resistor may be required for MISO to define the level when tri-stated
2. Decoupling capacitors must be placed close to the V_{DD} (Pin 4) and V_{DDIO} (Pin 5) supply pins of the ENS160

18 Soldering Information

The ENS160 uses an open LGA package. This package can be soldered using a standard reflow process in accordance with IPC/JEDEC J-STD-020D.

Figure 20: Solder Reflow Profile Graph



The detailed settings for the reflow profile are shown in the table below.

Table 38: Solder Reflow Profile

Parameter	Reference	Rate / Unit
Average temperature gradient in preheating		2.5K/s
Soak time	t_{SOAK}	2..3 min
Soak temp range	Ts max	200°C
	Ts min	150°C
Time above 217°C (T_1)	t_1	Max. 60s
Time above 230°C (T_2)	t_2	Max. 50s
Time above T_{PEAK} -10°C (T_3)	t_3	Max. 10s
Peak temperature in reflow	T_{PEAK}	260°C
Temperature gradient in cooling		Max. -5K/s

It is recommended to use a no-clean solder paste. There should not be any board wash processes, to prevent cleaning agents or other liquid materials contacting the sensor area.

19 Package Drawings & Markings

Figure 21: LGA Package Drawing

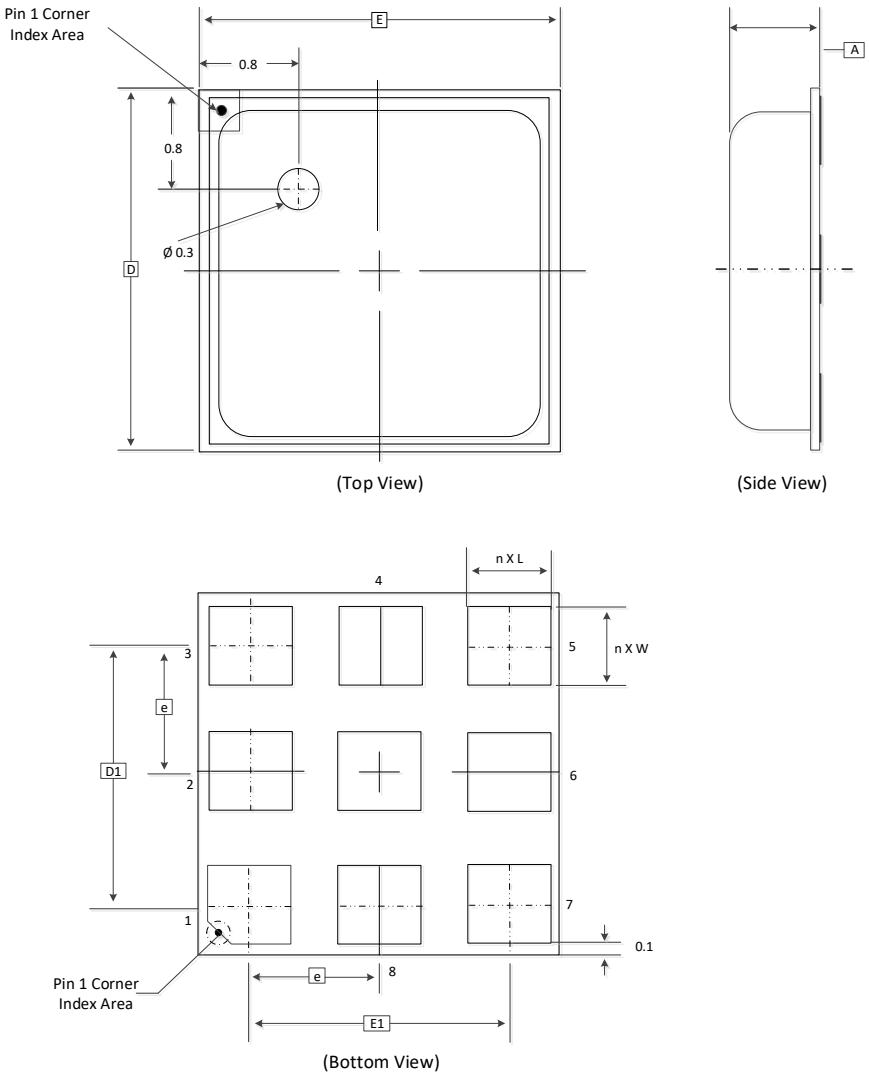
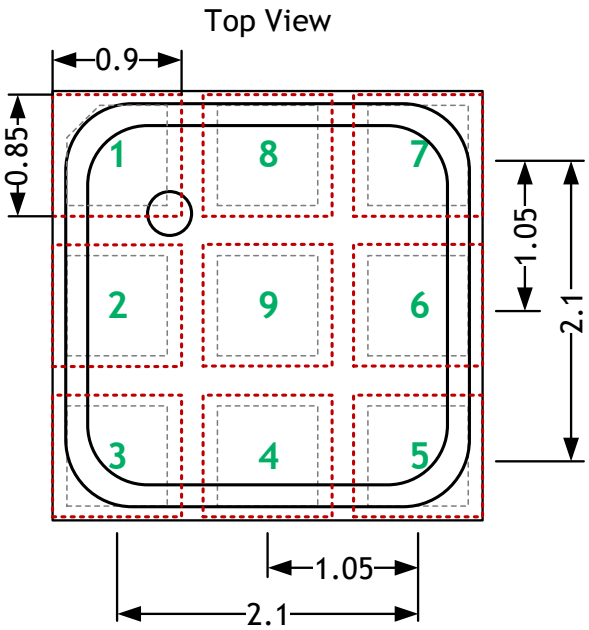


Table 39: LGA Package Dimensions

Parameter	Symbol	Dimensions		
		Min	Nominal	Max
Total thickness	A	-	0.83	0.9
Body Size	D		3.0	BSC
	E		3.0	BSC
Lead Width	W	0.65	0.7	0.75
Lead Length	L	0.65	0.7	0.75
Lead Pitch	e		1.05	BSC
Lead Count	n		9	
Edge Lead Centre to Centre	D1		2.1	BSC
	E1		2.1	BSC
Note: All dimensions are in mm				

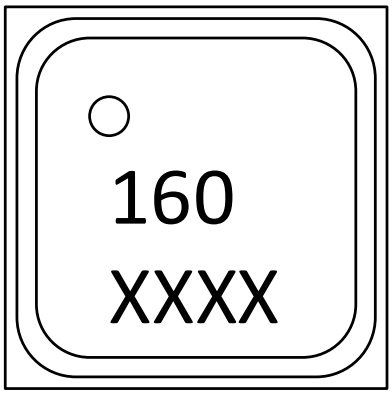
Figure 22: Recommend LGA Land Pattern for ENS160



Note(s):

1. All dimensions are in millimeters
2. PCB land pattern in dotted lines
3. Add 0.05mm all around the nominal lead width and length for the PCB land pattern

Figure 23: LGA Package Marking



20 RoHS Compliance & ScioSense Green Statement

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22 Document Status

Table 40: Document Status

Document Status	Product Status	Definition
Product Preview	Pre-Development	Information in this datasheet is based on product ideas in the planning phase of development. All specifications are design goals without any warranty and are subject to change without notice.
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23 Revision Information

Table 41: Revision History

Revision	Date	Comment	Page
0.95	2020-12-09	Preliminary Version – Product Launch	All
0.9	2019-12-11	Initial Version	All

Note(s) and/or Footnote(s):

1. Page and figure numbers for the previous version may differ from page and figure numbers in the current revision.
2. Correction of typographical errors is not explicitly mentioned.

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